

3) Explore AI and Machine Learning

Artificial Intelligence (AI) and Machine Learning (ML) are transforming the way we interact with technology. Understanding these concepts can position you at the forefront of innovation.

Understanding AI's role in daily life

Examples of AI in personal and professional tools

- **Personal Assistants:** Siri, Alexa, and Google Assistant use AI for voice recognition and responses.
- **Recommendation Systems:** Platforms like Netflix and Amazon suggest content based on AI algorithms.
- **Healthcare:** AI aids in diagnosing diseases and personalising treatment plans.

How AI is shaping industries and professions

- **Automation:** AI automates routine tasks, increasing efficiency.
- **Data Analysis:** Processes large datasets to identify patterns and insights.
- **Customer Service:** Chatbots provide immediate assistance.

Beginner-friendly Machine Learning resources

How to use platforms like Teachable Machine and ML for Kids

- **Teachable Machine:**
 - **Purpose:** An easy way to create machine learning models without coding.
 - **How It Works:**
 - **Train Models:** Use your webcam or upload images.
 - **Applications:** Recognize images, sounds, or poses.
- **ML for Kids:**
 - **Designed For:** Introducing children and beginners to machine learning concepts.
 - **Features:**
 - **Simple Projects:** Build games or applications using ML models.
 - **Visual Programming:** Uses Scratch for easy coding.

Introductory Courses on AI basics (e.g., Elements of AI)

- **Elements of AI:**
 - **Offered By:** University of Helsinki.
 - **Content:** Covers the basics of AI, its applications, and implications.
 - **Accessibility:** Free and designed for beginners.
- **Coursera and Udemy Courses:**
 - **Topics:** AI fundamentals, machine learning algorithms.
 - **Format:** Video lectures, quizzes, and projects.

Simple Machine Learning projects

Engaging in practical projects solidifies your understanding.

Building a Basic Predictive Model in Python

- **Objective:** Predict outcomes based on input data.
- **Steps:**
 1. **Install Libraries:** Use pip to install pandas, scikit-learn, and matplotlib.
 2. **Load Dataset:** Use datasets from scikit-learn or import your own.
 3. **Preprocess Data:** Clean and prepare data for modelling.
 4. **Choose a Model:** Start with linear regression or decision trees.
 5. **Train and Test:** Split data into training and testing sets.
 6. **Evaluate Performance:** Use metrics like accuracy or mean squared error.

Visualising patterns with Machine Learning tools

- **Use Cases:**
 - **Clustering:** Group similar data points.
 - **Classification:** Categorize data into predefined classes.
- **Tools:**
 - **Orange:** Visual programming software for data mining and ML.
 - **TensorFlow Playground:** Interactive visualisation of neural networks.

4) Understanding cloud computing and storage

Cloud computing has revolutionised how we store and access data. Understanding its basics is crucial for modern tech proficiency.

Overview of cloud basics

Difference between cloud storage and cloud computing

- **Cloud Storage:**
 - **Definition:** Storing data on remote servers accessed via the internet.
 - **Examples:** Google Drive, Dropbox, OneDrive.
 - **Benefits:**
 - **Accessibility:** Access files from any device.
 - **Scalability:** Easily increase storage capacity.
- **Cloud Computing:**
 - **Definition:** Delivery of computing services over the internet.
 - **Services Include:**
 - **Infrastructure as a Service (IaaS):** Virtualized computing resources.
 - **Platform as a Service (PaaS):** Platforms for developing, testing, and managing applications.
 - **Software as a Service (SaaS):** Software applications delivered over the internet.

Popular cloud providers and their services (AWS, Google Cloud, Azure)

- **Amazon Web Services (AWS):**
 - **Services:** Computing power, storage, databases, analytics.